Light

GCSE Computer Science — Networks Basics (Set 2)

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the primary function of a router in a computer network?

- ○ **A.** To forward data packets between different networks
- ○ **B.** To connect devices within the same network only
- ○ **C.** To store website data
- ○ **D.** To convert analogue signals to digital

Q2 (1 mark)

A school wants to connect 30 computers in one room so they can communicate with each other.
Which network device is most suitable?

- ○ **A.** A modem
- ○ **B.** A switch
- ○ **C.** A web server
- ○ **D.** A firewall

Q3 (1 mark)

What does NIC stand for in computer networking?

- ○ **A.** Network Internet Connection
- ○ **B.** Node Interconnect Cable
- ○ **C.** Network Interface Card
- ○ **D.** Network Installation Centre

Q4 (1 mark)

What is the purpose of a Wireless Access Point (WAP)?

- ○ **A.** To block unauthorised network traffic
- ○ **B.** To convert digital signals to analogue
- ○ **C.** To store network passwords
- ○ **D.** To allow wireless devices to connect to a wired network

Q5 (1 mark)

In a star network topology, how are devices connected?

- ○ **A.** Devices are connected in a circular loop
- ○ **B.** Each device connects directly to every other device
- ○ **C.** All devices connect to a central switch or hub
- ○ **D.** Devices are connected in a single line

Q6 (1 mark)

In a bus network topology, how is data transmitted?

- ○ **A.** Through a central server only
- ○ **B.** Along a single main cable that all devices share
- ○ **C.** In a circle from one device to the next
- ○ **D.** Wirelessly between all devices

Q7 (1 mark)

Which type of cable uses electrical signals to transmit data?

- ○ **A.** Fibre optic cable
- ○ **B.** Infrared cable
- ○ **C.** Copper cable (e.g., Ethernet)

- ○ **D.** Laser cable

Q8 (1 mark)

How does a fibre optic cable transmit data?

- ○ **A.** Using electrical signals through copper wires
- ○ **B.** Using radio waves through the air
- ○ **C.** Using magnetic pulses through iron cores
- ○ **D.** Using light pulses through glass or plastic fibres

Q9 (1 mark)

What is the difference between a hub and a switch?

- ○ **A.** A hub sends data only to the intended device; a switch broadcasts to all
- ○ **B.** A hub has no difference from a switch
- ○ **C.** A hub connects different networks; a switch connects devices in one network
- ○ **D.** A hub broadcasts data to all connected devices; a switch sends to specific devices

Q10 (1 mark)

A network connection has a bandwidth of 100 Mbps.
What does this mean?

- ○ **A.** The network has 100 connected devices
- ○ **B.** The network can transfer a maximum of 100 megabits per second
- ○ **C.** The network cable is 100 metres long
- ○ **D.** The network costs £100 per month

Q11 (1 mark)

In a full mesh network topology, how are devices connected?

- ○ **A.** All devices connect to one central device
- ○ **B.** Devices connect in a single line
- ○ **C.** Every device has a direct connection to every other device
- ○ **D.** Devices connect in a circular loop

Q12 (1 mark)

Which is a disadvantage of wireless networks compared to wired networks?

- ○ **A.** Wireless allows mobility
- ○ **B.** Wireless is easier to install
- ○ **C.** Wireless has no cables to manage
- ○ **D.** Wireless signals can be intercepted more easily

Q13 (2 marks)

Which is an advantage of a star topology over a bus topology?

- ○ **A.** Star uses less cabling than bus
- ○ **B.** Star has no single point of failure
- ○ **C.** Star is cheaper to set up than bus
- ○ **D.** If one cable fails in star, only that device is affected

Q14 (2 marks)

A company uses a bus topology for its network.
The main backbone cable develops a fault.
What happens to the network?

- ○ **A.** The entire network fails
- ○ **B.** Only devices near the fault are affected
- ○ **C.** The network automatically reroutes around the fault
- ○ **D.** Performance decreases but all devices stay connected

Q15 (2 marks)

A home user has a router provided by their Internet Service Provider (ISP).
Which function does this router perform?

- ○ **A.** Connects the home LAN to the ISP's wider network (the Internet)
- ○ **B.** Only provides wireless access within the home

- ○ **C.** Stores all the family's files and documents
- ○ **D.** Blocks all incoming internet traffic

Q16 (2 marks)

A business is choosing between copper and fibre optic cabling for their new office network.
Which statement correctly compares these options?

- ○ **A.** Copper is faster but more expensive than fibre
- ○ **B.** Fibre is immune to electromagnetic interference; copper is not
- ○ **C.** Copper can transmit data over longer distances than fibre
- ○ **D.** Fibre uses electrical signals; copper uses light

Q17 (2 marks)

Every NIC has a unique identifier burned into it during manufacture.
What is this identifier called?

- ○ **A.** IP address
- ○ **B.** URL
- ○ **C.** Domain name
- ○ **D.** MAC address

Q18 (2 marks)

Why might an organisation choose a mesh topology despite its higher cost?

- ○ **A.** High redundancy—multiple paths exist if one connection fails
- ○ **B.** It uses the least amount of cabling
- ○ **C.** It requires no network hardware
- ○ **D.** It is the easiest topology to set up

Q19 (2 marks)

A file of 80 megabits needs to be transferred over a network with a bandwidth of 10 Mbps.
What is the minimum time required to transfer this file?

- ○ **A.** 0.8 seconds
- ○ **B.** 8 seconds
- ○ **C.** 80 seconds
- ○ **D.** 800 seconds

Q20 (2 marks)

How does a switch know which port to send data to?

- ○ **A.** It guesses randomly
- ○ **B.** It broadcasts to all ports every time
- ○ **C.** It uses a MAC address table to match destinations to ports
- ○ **D.** It asks the user which port to use

Q21 (2 marks)

What is the main disadvantage of a star network topology?

- ○ **A.** Data collisions are very common
- ○ **B.** Adding new devices is very difficult
- ○ **C.** Only one device can be connected
- ○ **D.** If the central switch fails, the entire network fails

Q22 (2 marks)

What is the primary function of a modem?

- ○ **A.** To route data between different networks
- ○ **B.** To connect devices within a LAN
- ○ **C.** To convert digital signals to analogue and vice versa for transmission over telephone lines
- ○ **D.** To provide wireless network access

Q23 (2 marks)

Which statement about wireless transmission is correct?

- ○ **A.** Wireless always provides faster speeds than wired connections
- ○ **B.** Wireless signals cannot pass through any solid objects

- ○ **C.** Wireless connections are not affected by distance from the access point
- ○ **D.** Wireless uses radio waves to transmit data through the air

Q24 (3 marks)

A hospital needs a network for its critical patient monitoring systems. Which topology would be most appropriate and why?

- ○ **A.** Mesh, because multiple paths ensure data can still flow if one connection fails
- ○ **B.** Bus, because it uses the least cabling and is cheapest
- ○ **C.** Ring, because data travels quickly in one direction
- ○ **D.** Star with a single switch, because it's the easiest to manage

Q25 (3 marks)

A small business has these network requirements:
• 15 computers need to share files and a printer
• All devices need internet access
• Some employees use laptops and need wireless access
• The office has one floor with no physical barriers
Which combination of hardware would meet these requirements?

- ○ **A.** A hub and a modem only
- ○ **B.** A switch, a router, and a wireless access point
- ○ **C.** Only a wireless access point
- ○ **D.** A server and 15 NICs only

Q26 (3 marks)

A data centre is choosing cabling for connections that:
• Must carry data at very high speeds (100 Gbps)
• Will span distances of 500 metres
• Are near heavy electrical machinery
Which cable type is most suitable?

- ○ **A.** Standard copper Ethernet (Cat5e)
- ○ **B.** Coaxial cable
- ○ **C.** Fibre optic cable
- ○ **D.** Twisted pair telephone cable

Q27 (3 marks)

Compare these statements about network topologies:
1. Bus topology requires the least cabling
2. Star topology is easy to add new devices to
3. Mesh topology has no single point of failure
4. Bus topology provides high redundancy if one connection fails
Which statements are correct?

- ○ **A.** 1, 2, and 3 only
- ○ **B.** 1 and 4 only
- ○ **C.** 2, 3, and 4 only
- ○ **D.** All statements are correct

Q28 (3 marks)

A network administrator notices that when any device on the network sends data, all other devices receive it, causing slow performance.

Which hardware is most likely causing this problem?

- ○ **A.** The network is using a hub instead of a switch
- ○ **B.** The router is misconfigured
- ○ **C.** The fibre optic cables are damaged
- ○ **D.** The wireless access point has low signal strength

Q29 (3 marks)

A growing company currently has 20 employees but expects to have 200 within two years.
Which network design consideration is most important for their future needs?

- ○ **A.** Using the cheapest cabling available now
- ○ **B.** Scalability—choosing infrastructure that can grow with the company
- ○ **C.** Using bus topology because it's simplest
- ○ **D.** Avoiding all wireless connectivity

Q30 (3 marks)

A home router from an ISP typically combines multiple network devices in one unit.
Which combination of functions does a typical home router provide?

- ○ **A.** File server, print server, and email server
- ○ **B.** Modem, router, switch, and wireless access point
- ○ **C.** Firewall, antivirus, and backup storage

○ **D.** Web server, database server, and DNS server

Total: **55** marks • Time: **30 min**

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